SYLLABUS : MODERN PHYSICS & NUCLEAR PHYSICS

1. The mass number of a nucleus is

(A) always less than its atomic number

(B) always more than its atomic number

(C) equal to its atomic number

(D) sometimes more than and sometimes equal to its atomic number

2. The stable nucleus that has a radius 1/3 that of Os^{189} is -

(A) $_3Li^7$ (B) $_2He^4$ (C) $_5B^{10}$ (D) $_6C^{12}$

3. The mass of proton is 1.0073 u and that of neutron is 1.0087 u (u = atomic mass unit).

The binding energy of $_2^4He$ is :– [Given helium nucleus mass $\approx$4.0015u]

(A) 0.0305 J (B) 0.0305 erg (C) 28.4 MeV (D) 0.061 u

4. The binding energy of deuteron is 2.2 MeV and that of $_2^4He$ is 28 MeV. If two deuterons are fused to form one ^4_2He then the energy released is :–

(A) 23.6 MeV (B) 19.2 MeV (C) 30.2 MeV (D) 25.8 MeV

5. As the mass number A increases, the binding energy per nucleon in a nucleus

(A) increases

(B) decreases

(C) remains the same

(D) varies in a way that depends on the actual value of A.

6. Which of the following is a wrong description of binding energy of a nucleus ?

(A) It is the energy required to break a nucleus into its constituent nucleons.

(B) It is the energy released when free nucleons combine to from a nucleus

(C) It is the sum of the rest mass energies of its nucleons minus the rest mass energy of the nucleus

(D) It is the sum of the kinetic energy of all the nucleons in the nucleus

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7. The energy of the reaction $Li^7 + p \longrightarrow 2\,He^4$ is (the binding energy per nucleon in Li^7 and He^4 nuclei are 5.60 and 7.06 MeV respectively.)

(A) 17.3 MeV (B) 1.73 MeV

(C) 1.46 MeV (D) depends on binding energy of proton

8. The atomic weight of boron is 10.81 g/mole and it has two isotopes $^{10}_{5}B$ and $^{11}_{5}B$. The ratio (by number) of $^{10}_{5}B$: $^{11}_{5}B$ in nature would be :

(A) 19 : 81 (B) 10 : 11 (C) 15 : 16 (D) 81 : 19

9. An α-particle is bombarded on ^{14}N. As a result, a ^{17}O nucleus is formed and a particle is emitted. This particle is a

(A) neutron (B) proton (C) electron (D) positron

10. Which of the following radiations has the least wavelength?

(A) γ-rays (B) β-rays (C) α-rays (D) X-rays

11. The binding energy per nucleon of deuteron $\left(^{2}_{1}H\right)$ and helium nucleus $\left(^{4}_{2}He\right)$ is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is :

(A) 13.9 MeV (B) 26.9 MeV (C) 23.6 MeV (D) 19.2 MeV

12. If radius of the $^{27}_{13}Al$ nucleus is estimated to be 3.6 Fermi, then the radius of $^{125}_{52}Te$ nucleus be nearly:

(A) 6 Fermi (B) 8 Fermi (C) 4 Fermi (D) 5 Fermi

13. If the binding energy per nucleon in $^{7}_{3}Li$ and $^{4}_{2}He$ nuclei are 5.60 MeV and 7.06 MeV respectively, then in the reaction

$$p + {}^{7}_{3}Li \rightarrow 2\,{}^{4}_{2}He$$

energy of proton must be :

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(A) 39.2 MeV (B) 28.24 MeV (C) 17.28 MeV (D) 1.46 MeV

14. If M_o is the mass of an oxygen isotope $_8O^{17}$, M_p and M_N are the masses of a proton and a neutron respectively, the nuclear binding energy of the isotope is :

(A) $(M_o - 8M_P)C^2$ (B) $(M_o - 8M_P - 9M_N)C^2$

(C) M_oC^2 (D) $(M_o - 17M_N)C^2$

15. This binding energy per nucleon for the parent nucleus is E_1 and that for the daughter nuclei is E_2. Then:

(A) $E_1 = 2E_2$ (B) $E_1 > E_2$ (C) $E_2 > E_1$ (D) $E_2 = 2E_1$

16. Consider the nuclear fission

$Ne^{20} \rightarrow 2He^4 + C^{12}$

Given that the binding energy/nucleon of Ne^{20}, He^4 and C^{12} are, respectively, 8.03 MeV, 7.07 MeV and 7.86 MeV, identify the correct statement.

(A) Energy of 9.72 MeV of energy will be supplied. (B) 8.3 MeV energy will be released

(C) Energy of 3.6 MeV will be released (D) Energy of 11.9 MeV has to be supplied

17. Given the masses of various atomic particles m_p = 1.0072 u, m_n = 1.0087 u, m_e = 0.000548 u, $m_{\nu}^{-} = 0$, m_d = 2.0141 u, where p ≡ proton, n ≡ neutron, e ≡ electron, $\bar{\nu}$ ≡ antineutrino and d ≡ deuteron. Which of the following process is allowed by momentum and energy conservation?

(A) n + n → deuterium atom (electron bound to the nucleus)

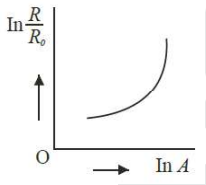
(B) n + p → d + γ (C) $p \rightarrow n + e^+ + \bar{\nu}$ (D) $e^+ + e^- \rightarrow \gamma$

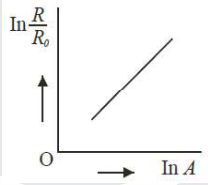
18. Nucleus A is having mass number 220 and its binding energy per nucleon is 5.6 MeV. It splits in two fragments 'B' and 'C' of mass numbers 105 and 115. The binding energy of nucleons in 'B' and 'C' is 6.4 MeV per nucleon. The energy Q released per fission will be :

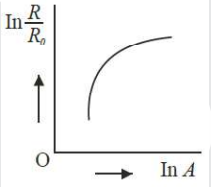
(A) 0.8 MeV (B) 275 MeV (C) 220 MeV (D) 176 MeV

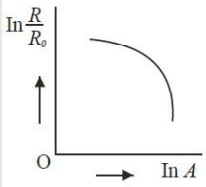
19. Which of the following figure represents the variation of $\ln\left(\frac{R}{R_0}\right)$ with ln A(If R = radius of a nucleus and

A = its mass number)

(A) 

(B) 

(C) 

(D) 

20. Read the following statements :

(a) Volume of the nucleus is directly proportional to the mass number.

(b) volume o the nucleus is independent of mass number.

(c) Density of the nucleus is directly proportional to the mass number.

(d) Density of the nucleus is directly proportional to the cube root of the mass number.

(e) Density of the nucleus is independent of the mass number.

Choose the correct option from the following options.

(A) (a) and (d) only (B) (a) and (e) only (C) (b) and (e) only (D) (a) and (c) only

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21. Per nucleon energy of $_3Li^7$ and $_2He^4$ nucleus is 5.60 MeV and 7.06 MeV, then in $_3Li^7 + {}_1P^1 \rightarrow 2_2He^4$ energy released is :–

(A) 29.6 MeV (B) 2.4 MeV (C) 8.4 MeV (D) 17.3 MeV

22. From the given data, the amount of energy required break the nucleus of aluminium $^{27}_{13}Al$ is _____ x × 10^{-3} J.

Mass of neutron = 1.00566 u

Mass of proton = 1.00726 u

Mass of Aluminium nucleus = 27.18846 u

(Assume 1 u corresponds to x J of energy)

(Round off to the nearest integer)

(A) 21 (B) 25 (C) 27 (D) 30

23. $_{92}U^{235}$ nucleus absorbs a slow neutron and undergoes fission into $_{54}X^{139}$ and $_{38}Sr^{94}$ nuclei. The other particles produced in this fission process are

(A) 1 β and 1 α (B) 2 β and 1 neutron (C) 2 neturons (D) 3 neutrons

24. In a fission reaction

$$^{236}_{92}U \longrightarrow {}^{117}X + {}^{117}Y + n + n$$

the average binding energy per nucleon of X and Y is 8.5 MeV whereas that of ^{236}U is 7.6 MeV. The total energy liberated will be about :

(A) 200 keV (B) 2 MeV (C) 200 MeV (D) 2000 MeV

25. A heavy nucleus having mass number 200 gets disintegrated into two small fragments of mass number 80 and 120. If binding energy per nucleon for parent atom is 6.5 M eV and for daughter nuclei is 7 M eV and 8 M eV respectively, then the energy released in each decay will be :

(A) 200 M eV (B) – 220 M eV (C) 220 M eV (D) 180 M eV

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ANSWER KEY

1.	(D)	**2.**	(A)	**3.**	(C)	**4.**	(A)	**5.**	(D)
6.	(D)	**7.**	(A)	**8.**	(A)	**9.**	(B)	**10.**	(A)
11.	(C)	**12.**	(A)	**13.**	(C)	**14.**	(B)	**15.**	(C)
16.	(A)	**17.**	(B)	**18.**	(D)	**19.**	(B)	**20.**	(B)
21.	(D)	**22.**	(C)	**23.**	(D)	**24.**	(C)	**25.**	(C)